

The branch choice in L -functions of weakly holomorphic modular forms, read as a choice of $(1 - T)^{-1}$

Working note, project “Hecke and MFs with poles” (drafted with Claude)

24 September 2026

Abstract

For $f \in M_k^!$ with a pole at the cusp, the L -series of Bringmann–Fricke–Kent contains, for each polar Fourier mode $c(-m)q^{-m}$, incomplete gamma functions $\Gamma(s, z)$ at the negative real argument $z = -2\pi m$. That point lies on the branch cut, so the series implicitly requires a choice of side. Diamantis–Lee–Raji–Rolen fix the principal branch. This note makes the choice explicit and says what it means. The finite-contour L -integral of the project draft uses a Hurwitz T -kernel, which solves a shift-difference equation, i.e. inverts $1 - T$. The two one-sided inversions $h_{\pm}$ unfold into horizontal Mellin rays inside $\mathbb{H}$, one running to the right and one to the left. They reproduce exactly the two sides of the cut, and h_+ is the principal branch. Their difference is a residue at $q = 0$:

$$L_+^*(f, s) - L_-^*(f, s) = -2\pi i \left[\frac{D(f, s)}{(2\pi)^s \Gamma(1-s)} + i^k \frac{D(f, k-s)}{(2\pi)^{k-s} \Gamma(1-k+s)} \right], \quad D(f, s) = \sum_{m \geq 1} c_f(-m) m^{-s}.$$

This is *not* a new term. It is the sum of the per-mode jumps across the cut. The two sides are complex conjugates at real s , so their mean is real (a principal value). The note contributes three things: an explicit account of a choice that is usually made silently; a dictionary identifying the side of the cut of $\Gamma(s, \cdot)$ with a one-sided coboundary inversion of $1 - T$ and with the direction of a horizontal Mellin ray; and consequences of that dictionary. The choice is invisible on M_k and at the critical integers. The functional equation forces the same side at both cusps. The principal-branch values are not real on the real axis. Every step takes place inside $\mathbb{H}$, on individual Fourier modes, or at the isolated point $q = 0$; f is never continued across the real axis.

1 Framing, and the natural-boundary concern

What this note is, and is not. Passing from decaying Fourier modes ($n > 0$) to growing ones ($n < 0$) in the incomplete-gamma representation is an analytic continuation in the argument of $\Gamma(s, \cdot)$. That continuation lands on the cut, so it involves a choice, and that choice is the entire content of the ambiguity studied here. We do not claim a new term or a new invariant. We claim a precise account of the choice (Theorem 6, Proposition 9) and an identification of it with the coboundary freedom in inverting $1 - T$. That second freedom is the one that already appears on the cocycle side when period polynomials are built from the same kernel at integer s .

The natural-boundary concern. A modular form lives on $\mathbb{H}$. The real line is a natural boundary: every rational point is a cusp, and for f with a pole at $i\infty$ the modular transformation law makes f grow exponentially along every geodesic ending at a rational. So a phrase like

“rotate the Mellin contour of f into the lower half-plane” has no meaning. In an earlier informal explanation we used exactly that phrase. It is only correct for *individual Fourier modes* $q^n = e^{2\pi i n \tau}$, which are entire in τ . This note replaces that heuristic with statements that stay inside the domain of f :

1. The two kernels unfold into two horizontal Mellin rays inside $\mathbb{H}$ (Lemma 3).
2. For f with a pole at the cusp, horizontal is the only admissible direction for a ray, so there are exactly two choices (Lemma 4).
3. No deformation inside $\mathbb{H}$ connects the two rays. Going over the top is blocked by the pole at $q = 0$, and going underneath is blocked by the natural boundary. The obstruction is measured *exactly* by the residue at $q = 0$ (Theorem 6). In the q -coordinate, the cusp $i\infty$ is an isolated singular point, while every other cusp lies on the circle $|q| = 1$.

The lower half-plane and the Stokes-type branch jump of $\Gamma(s, z)$ return only in Section 6, as a computation carried out on finitely many entire Fourier modes.

2 Setup

Let k be even and $f \in M_k^!$ (no interior poles; see Remark 13), with $f(\tau) = \sum_{n \geq -M} c(n)q^n$. Let $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Powers τ^w use the principal branch on $\mathbb{C} \setminus (-\infty, 0] \supset \mathbb{H}$. For $\tau \in \mathbb{H}$ we have $\arg(-1/\tau) = \pi - \arg \tau$. Hence, with the weight $2 - k$ action $(g|_{2-k}\gamma)(\tau) = (c\tau + d)^{k-2}g(\gamma\tau)$ and $\varphi_s(\tau) = \tau^{s-1}$,

$$\varphi_s|_{2-k}S = e^{i\pi(s-1)}\tau^{k-1-s} \quad \text{exactly on } \mathbb{H}. \quad (1)$$

Definition 1 (One-sided inversions of $1 - T$). For $\tau \in \mathbb{H}$ and $\operatorname{Re} w < -1$ put

$$h_+[w](\tau) = \sum_{n \geq 1} (\tau + n)^w = \zeta(-w, \tau + 1), \quad h_-[w](\tau) = - \sum_{n \leq 0} (\tau + n)^w = -e^{i\pi w} \zeta(-w, -\tau),$$

and continue meromorphically in w . (In the closed form of h_- the Hurwitz argument $-\tau$ lies in the lower half-plane. This is an evaluation of the elementary function $\zeta(-w, \cdot)$ only; f is never evaluated off $\mathbb{H}$ anywhere in this note, except for individual Fourier modes in Section 6.) Both satisfy $h[w](\tau) - h[w](\tau - 1) = -\tau^w$. Define $K_\pm(\tau, s) = h_\pm[s - 1](\tau) - e^{i\pi(s-1)}h_\pm[k - 1 - s](\tau)$, so that $K_\pm(\tau) - K_\pm(\tau - 1) = -(\varphi_s|_{2-k}(1 - S))(\tau)$. The kernel K_+ is the Hurwitz T -kernel k_T of the draft.

Definition 2 (Finite-contour L -integral). For $\tau_0 \in \mathbb{H}$, $\gamma^S : -1/\tau_0 \rightarrow \tau_0$ and $\gamma^T : \tau_0 - 1 \rightarrow \tau_0$,

$$L_\pm^*(f, s) = e^{-\pi i s/2} \left(\int_{\gamma^S} f(\tau) \tau^{s-1} d\tau + \int_{\gamma^T} f(\tau) K_\pm(\tau, s) d\tau \right).$$

Differentiating in τ_0 with $f(-1/\tau) = \tau^k f(\tau)$ and (1), the S -segment contributes $f(\tau_0)(\varphi_s|(1 - S))(\tau_0)$ and the T -segment contributes $f(\tau_0)(K(\tau_0) - K(\tau_0 - 1))$. These cancel for *any* solution K of the shift equation. So basepoint independence cannot select a kernel. The solutions form $K_+ + \{1\text{-periodic functions}\}$.

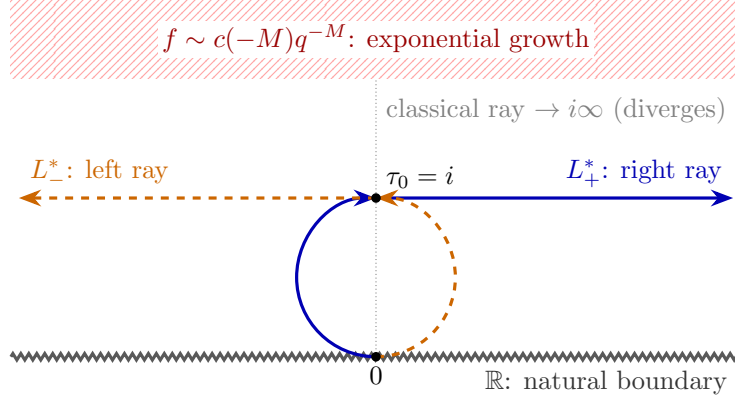


Figure 1: The two unfolded Mellin paths for $\tau_0 = i$ (where γ^S degenerates). Solid: L_+^* enters the cusp 0 along a horocycle from the left and leaves toward ∞ horizontally to the right. Dashed: L_-^* , the mirror image. Neither path can be deformed into the other: the region above (the cusp $i\infty$) and the real axis below are both forbidden.

3 Unfolding: the kernels are horizontal Mellin rays

Lemma 3. *Let γ^T be the horizontal segment at height $y_0 = \text{Im } \tau_0$ and $\text{Re } w < -1$. Then*

$$\int_{\tau_0-1}^{\tau_0} f h_+[w] d\tau = \int_0^\infty f(\tau_0+t) (\tau_0+t)^w dt, \quad \int_{\tau_0-1}^{\tau_0} f h_-[w] d\tau = - \int_{-\infty}^0 f(\tau_0+t) (\tau_0+t)^w dt,$$

with $t \in \mathbb{R}$. Both right-hand integrals therefore run along the horizontal line $\text{Im } \tau = y_0 > 0$, to the right and to the left respectively, and converge absolutely. Along the left ray $\arg \tau \rightarrow \pi$ but stays below π , so τ^w never meets its cut. The “endpoint at infinity” is the cusp $\infty \in \mathbb{P}^1$, approached at fixed height.

Proof. On the line, f is continuous and 1-periodic, hence bounded. Since $|\tau^w| \leq C|\tau|^{\text{Re } w}$, the sum $\sum_n$ converges uniformly on the segment, and we may exchange it with the integral. Periodicity gives $\int_{\tau_0-1}^{\tau_0} f(\tau)(\tau+n)^w d\tau = \int_{\tau_0-1+n}^{\tau_0+n} f(u)u^w du$. Summing over $n \geq 1$ tiles the right ray; summing over $n \leq 0$ tiles the left ray. $\square$

So for $\text{Re } s < 0$ the first half of K_+ is the Mellin integral of f along the horizontal ray from τ_0 to the right. The second half, valid for $\text{Re } s > k$, becomes the same kind of ray integral against τ^{k-1-s} . The substitution $u = -1/\tau$ turns it into a Mellin integral of $f u^{s-1}$ along the S -image of the ray. That image is an arc of the horocycle $|u - \frac{i}{2y_0}| = \frac{1}{2y_0}$, which approaches the cusp 0 tangentially from the left, since $u = -1/(t + iy_0)$ has $\text{Re } u < 0$.

Put together, and with each piece continued separately in s , L_+^* is the Mellin integral of f along a path from the cusp 0 to the cusp ∞ . It follows the classical route, but enters and leaves both cusps *horocyclically*, not along the geodesic $i\mathbb{R}_{>0}$ (Figure 1). L_-^* is its mirror image under $\tau \mapsto -\bar{\tau}$.

4 Why exactly two: admissible directions and the two walls

Lemma 4. *Let $M \geq 1$ and $c(-M) \neq 0$. For the ray $\rho_\theta(t) = \tau_0 + te^{i\theta}$, $t \geq 0$:*

(a) if $0 < \theta < \pi$, then $\int_{\rho_\theta} f(\tau) \tau^w d\tau$ diverges for every $w \in \mathbb{C}$;

(b) if $\theta \in \{0, \pi\}$, it converges absolutely for $\operatorname{Re} w < -1$;

(c) if $\pi < \theta < 2\pi$, the ray reaches $\mathbb{R}$ at finite t , where f is not defined.

Proof. (a) As $\operatorname{Im} \tau \rightarrow \infty$ uniformly in $\operatorname{Re} \tau$, we have $f(\tau) = c(-M)q^{-M}(1 + O(e^{-2\pi \operatorname{Im} \tau}))$. Along ρ_θ , $|q^{-M}| = e^{2\pi M(y_0 + t \sin \theta)}$ grows exponentially, while $|\tau^w|$ changes only polynomially. The partial integrals therefore grow exponentially. (b) is Lemma 3. (c) holds because the ray leaves $\mathbb{H}$. $\square$

The natural boundary. For $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \operatorname{SL}_2(\mathbb{Z})$ with $c \neq 0$, $f(\gamma\tau) = (c\tau + d)^k f(\tau)$. As $\tau \rightarrow i\infty$, $\gamma\tau$ tends to a/c along a geodesic, and $|f(\gamma\tau)|$ grows like $e^{2\pi M \operatorname{Im} \tau}$. So f blows up exponentially at a dense set of real points and has no holomorphic continuation across any real interval. The following bookkeeping records what may legitimately be done with each object.

object	domain of holomorphy	allowed operations
f	$\mathbb{H}$ (in q : $0 < q < 1$, pole of order M at 0)	deform contours inside $\mathbb{H}$ only
single mode q^n $\tau^w, h_\pm[w]$	all of $\mathbb{C}$ $\mathbb{C}$ minus a cut	deform anywhere (mind the cut of τ^w) continue freely, but never pair with f off $\mathbb{H}$
$F(q) = f(\tau)$	$0 < q < 1$; $ q = 1$ is a natural boundary	residues at $q = 0$

Consequence. For a cusp form ($M = 0, c(0) = 0$), f decays at $i\infty$, and every ray with $\theta \in [0, \pi]$ gives the same value. Deforming through the region near the cusp is legitimate, because the arcs at infinity vanish. For $M \geq 1$ that region is forbidden, and the real axis is forbidden for any f . The two admissible rays are separated above by the cusp and below by the natural boundary. Their difference is therefore a well-defined obstruction and need not vanish. We compute it next.

5 The difference is a residue at the cusp

Lemma 5 (Lipschitz). *For $\tau \in \mathbb{H}$ and $\operatorname{Re} w < -1$,*

$$\Lambda_w(\tau) := \sum_{n \in \mathbb{Z}} (\tau + n)^w = \frac{(-2\pi i)^{-w}}{\Gamma(-w)} \sum_{r \geq 1} r^{-w-1} q^r.$$

In particular

$$h_+[s-1] - h_-[s-1] = \Lambda_{s-1} = \frac{(-2\pi i)^{1-s}}{\Gamma(1-s)} \operatorname{Li}_s(q),$$

and this identity continues meromorphically in s .

Λ_w is 1-periodic, so it descends to the q -disc, where it is holomorphic and vanishes at $q = 0$. It is precisely a “periodic correction” of the kernel.

Theorem 6. *For $f \in M_k^1$ and all s (as an identity of meromorphic functions),*

$$L_+^*(f, s) - L_-^*(f, s) = -2\pi i \left[\frac{D(f, s)}{(2\pi)^s \Gamma(1-s)} + i^k \frac{D(f, k-s)}{(2\pi)^{k-s} \Gamma(1-k+s)} \right], \quad D(f, s) = \sum_{m=1}^M \frac{c(-m)}{m^s}.$$

Proof. First, $L_+^* - L_-^* = e^{-\pi i s/2} \int_{\gamma_T} f [\Lambda_{s-1} - e^{i\pi(s-1)} \Lambda_{k-1-s}] d\tau$. By Lemma 3, $\int_{\gamma_T} f \Lambda_w$ equals the Mellin integral of f over the *full* horizontal line: right ray minus left ray. To evaluate it, put $q = e^{2\pi i \tau}$. The segment becomes the circle $|q| = e^{-2\pi y_0}$, traversed once counter-clockwise, and $d\tau = dq/(2\pi i q)$. Then

$$\int_{\tau_0-1}^{\tau_0} f \Lambda_w d\tau = \oint \frac{F(q) \Lambda_w(q)}{2\pi i q} dq = \operatorname{Res}_{q=0} \frac{F \Lambda_w}{q} = \frac{(-2\pi i)^{-w}}{\Gamma(-w)} \sum_{m \geq 1} c(-m) m^{-w-1}.$$

The only singularity of $F \Lambda_w / q$ inside the circle is at $q = 0$; the natural boundary $|q| = 1$ is never approached. Taking $w = s - 1$ and $w = k - 1 - s$ and simplifying the phases with (1) gives the claim. $\square$

Corollary 7. (i) $L_+^* = L_-^*$ on M_k . The constant term is unambiguous too, since Λ_w has no q^0 term.

(ii) $L_+^* = L_-^*$ at $s \in \{1, \dots, k-1\}$, because $1/\Gamma$ vanishes there. Periods and critical values are canonical.

(iii) The two terms are the choices made at the two cusp ends of the path in Figure 1. The term $D(f, s)$ belongs to ∞ , and the term $D(f, k-s)$ belongs to 0. The data at 0 is the same principal part, transported by S . The functional equation $L^*(s) = i^k L^*(k-s)$ holds for K_+ , for K_- , and for every affine combination $\lambda K_+ + (1-\lambda)K_-$. It fails for the mixed kernel that uses h_+ in one half and h_- in the other (Table 1). So the functional equation forces the path to be S -symmetric.

Remark 8 (What distinguishes “right” from “left”). In the q -disc, both horizontal rays wind infinitely often around the same circle and never approach $q = 0$. The distinction is invisible on $Y(1)$ and exists only on the universal cover $\mathbb{H}$. It arises because the Mellin weight τ^{s-1} is not T -invariant, so one must decide in which direction to sum the T -orbit. The cusp of $Y(1)$ has no “sides”. The two sides are introduced by the non-invariant kernel, and the residue at $q = 0$ measures how much f notices. Cohomologically, the ambiguity is $\ker(1-T)$ on the kernel module. On the polynomial module V_{k-2} (integer s) this kernel consists of constants, which pair with $c(0)$ only. At non-integer s it contains $\operatorname{Li}_s(q)$, which pairs with the principal part.

6 The per-mode shadow: where the lower half-plane legitimately enters

The Bringmann–Fricke–Kent (BFK) formula at $\tau_0 = i$ is

$$L^*(f, s) = \sum_{n \neq 0} c(n) \left[\frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right] - c(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right).$$

It arises from integrating each Fourier mode separately. On the horizontal ray this is rigorous: the Fourier series converges uniformly there, and by the bound in the proof of Lemma 3 we may exchange sum and integral. Each mode is now entire, so its own contour may be rotated. For $n > 0$ we rotate upward to $i\infty$, which reproduces the classical incomplete gamma. The principal part has only finitely many modes q^{-m} , and each decays in the *lower* half-plane. So for these modes alone we rotate the horizontal ray downward. This is legitimate for $q^{-m} \tau^w$ and meaningless

for f . The right ray, rotated clockwise, passes the branch point $\tau = 0$ of τ^w on the right, inside the cut plane. The left ray passes it on the left and lands on the other sheet. The two results differ by a Hankel loop around $\tau = 0$. The Hankel loop gives $(-2\pi i)^{-w} m^{-w-1}/\Gamma(-w)$, which is the q^m coefficient of Λ_w , in agreement with Theorem 6.

Two remarks. First, the branch point of the kernel sits at $\tau = 0$, which is itself a cusp: the S -image of $i\infty$. The per-mode computation sees a branch point exactly where f has a natural-boundary point. The per-mode and global pictures agree only because the sum over modes is taken *on the horizontal ray, before any rotation*. Summing rotated mode integrals over infinitely many modes and calling the result an integral of f in the lower half-plane would be meaningless. Second, in BFK's formula the negative- n terms are values of $z^{-s}\Gamma(s, z)$ on its cut $z < 0$. Writing $z^{-s}\Gamma(s, z) = \Gamma(s)z^{-s} - E$, with

$$E(s, m) := \sum_{j \geq 0} \frac{(2\pi m)^j}{j!(s+j)} \quad (z = -2\pi m),$$

single-valued, only $\Gamma(s)z^{-s}$ depends on the branch. Its jump, $\Gamma(s)(2\pi m)^{-s}(e^{-i\pi s} - e^{i\pi s}) = -2\pi i(2\pi m)^{-s}/\Gamma(1-s)$, is exactly the term in Theorem 6. This yields the following statement.

Proposition 9 (Global difference = summed per-mode jumps). *Let $B_{\pm}(s, m)$ denote the polar-mode summand $\frac{\Gamma(s, z)}{z^s} + i^k \frac{\Gamma(k-s, z)}{z^{k-s}}$ evaluated as the limit $z \rightarrow -2\pi m \pm i0$. Then*

$$L_+^*(f, s) - L_-^*(f, s) = \sum_{m \geq 1} c(-m) [B_+(s, m) - B_-(s, m)],$$

and L_+^* equals the BFK series with every polar term taken from the upper side ($\arg z = +\pi$, the principal branch; this is the convention made explicit in [2]). L_-^* takes every polar term from the lower side.

Proof. The jump of $z^{-a}\Gamma(a, z)$ across $z = -x$ is $-2\pi i x^{-a}/\Gamma(1-a)$. Summing this over both halves and all polar modes reproduces the right-hand side of Theorem 6. $\square$

This was checked independently of the closed-form jump. `mpmath`'s own continuation of $\Gamma(s, z)$ at $z = -2\pi m \pm 10^{-30}i$ reproduces the kernel difference to 10^{-26} or better (Table 1). The global Dirichlet-polynomial formula and the per-mode branch picture are therefore the same ambiguity, seen at two scales. The link between them is the Lipschitz formula. This is a computed identity, not something visible from the shape of the formula.

Remark 10 (The branch choice is a special case of a larger freedom). The shift-difference equation allows any 1-periodic correction $p(\tau, s)$ of the kernel. If p is holomorphic at $q = 0$, with $p = \sum_{n \geq 0} p_n(s)q^n$, then L^* changes by $e^{-\pi i s/2} \sum_{m \geq 0} c(-m)p_m(s)$. For instance, $p = q$ shifts L^* by $e^{-\pi i s/2} c(-1)$, which is not a branch effect of anything. The functional equation only requires $e^{-\pi i s/2} p_m(s)$ to be symmetric under $s \leftrightarrow k-s$, which still leaves infinitely many choices. The branch choice is the one-parameter subfamily obtained by inverting $1-T$ as a one-sided sum over a T -orbit of τ^{s-1} , together with affine combinations of the two. Any principle that singles out these one-sided sums is extra input; it does not follow from basepoint independence or from the functional equation.

Branch-explicit formula. Averaging the two kernels gives, with $n = -m < 0$,

$$L_{\text{sym}}^*(f, s) = \sum_{n>0} c(n) \left[\frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right] - c(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right) \\ + \sum_{m \geq 1} c(-m) \left[\cos(\pi s) (\Gamma(s) (2\pi m)^{-s} + i^k \Gamma(k-s) (2\pi m)^{-(k-s)}) - E(s, m) - i^k E(k-s, m) \right],$$

$$\text{and } L_{\pm}^* = L_{\text{sym}}^* \mp i\pi \left[\frac{D(f, s)}{(2\pi)^s \Gamma(1-s)} + i^k \frac{D(f, k-s)}{(2\pi)^{k-s} \Gamma(1-k+s)} \right].$$

7 Reality: generic behaviour of a cut, and what it costs

For real s and real $x > 0$, the two lateral values of $z^{-s}\Gamma(s, z)$ at $z = -x$ are complex conjugates. This is the generic behaviour of a real-analytic function on its cut, and it has nothing to do with modular forms. Their mean is real, and it is the natural principal value. For $s = 0$ this is literal: the two sides of $\Gamma(0, -x)$ are $-\text{Ei}(x) \mp i\pi$, and $\text{Ei}(x)$ is defined by a Cauchy principal-value integral (checked at $x = 2\pi$ to 40 digits). So the reality of $L_{\text{sym}}^* = \frac{1}{2}(L_+^* + L_-^*)$ is not a discovery. It is the statement that one takes the principal value on the cut.

Geometrically, the map $\tau \mapsto -\bar{\tau}$ normalises $\text{SL}_2(\mathbb{Z})$, conjugates T to T^{-1} , and exchanges the two horizontal rays. For f with real coefficients, $f(-\bar{\tau}) = f(\tau)$, and $L_-^*(f, s) = \overline{L_+^*(f, s)}$ for real s (checked numerically).

What does deserve to be recorded is the cost of the standard convention. When $D(f, \cdot) \not\equiv 0$, the principal-branch L -values (equivalently L_+^*) are *not real* on the real axis, even for f with real coefficients. Anyone importing the intuition from M_k , where real coefficients give real values at real s , would be misled. The imaginary part is exactly half of the correction in Theorem 6. There is a formal resemblance to lateral versus median Borel resummation: two admissible lateral directions, a forbidden direction (here, toward the cusp), and an imaginary ambiguity. The mechanism here, however, is a residue at a pole in the q -disc, not a Borel-plane singularity.

8 Relation to the period theory of the project draft

The longer project draft attaches periods and quasi-periods to meromorphic forms. As described there (the draft was not re-read for this note, so the details below are the author's framing, not independently checked), two ambiguities arise:

- (a) *wall-crossing*: a contour crossing an interior pole shifts every special value by a residue;
- (b) *subtraction*: the regularising subtraction is defined only modulo $S_k^!$, a space of dimension $2 \dim S_k$.

For $\dim S_k = 1$ ($k = 12, 16, 18, 20, 22, 26$), the periods of $\Delta_k = \Delta E_{k-12}$ and the quasi-periods of $\hat{\Delta}_k = q^{-1} + O(q)$ span the $W^+ \oplus W^-$ plane. The subtraction coefficients are arbitrary complex numbers, so there is no lattice. Modulo subtraction, the period vector is a smear that carries no information. The substantive question is whether a *canonical section* exists: a canonical rule for the subtraction. The choice between K_+ and K_- studied here is a small instance of the same question.

Remark 11 (The branch choice is transverse to the subtraction smear; reasoned, unchecked). By Corollary 7(ii), $L_+^* - L_-^*$ vanishes at every $s \in \{1, \dots, k-1\}$ (checked directly at $s = 3, 6, 9$ for

Δ' and g). It therefore does not touch the V_{k-2} period polynomial or its $W^\pm$ coordinates. It is visible only at non-critical s , including $s = 0$ and $s = k$, where it is nonzero. This assumes the draft builds periods from $s = 1, \dots, k-1$ only.

Remark 12 (Wall-crossing jumps are subtraction-invariant; reasoned, unchecked). Subtracting a weakly holomorphic form adds no residues in $\mathbb{H}$. Jumps across walls, and hence differences of periods between homotopy classes of contours, are therefore unaffected by the subtraction ambiguity, even though absolute periods are. If canonical content survives the smear, relative periods are a natural place to look. A form with a pole at a CM point (i or ρ), where the Laurent data is algebraic, is the obvious test case.

9 Numerical checks

All checks are reproduced by `verify_kernel.py`, which reports 47/47 PASS with `--slow`. They use `mpmath` at 30 digits and $\tau_0 = i$. The test forms are $\Delta' = \frac{E_6^2 E_4^3}{\Delta} + \frac{5541}{16} E_4^3 - \frac{1317}{16} E_6^2 = q^{-1} + 47709536q^2 + \dots$ (Brown [3]), and $g = j\Delta' - 744\Delta' = q^{-2} + 196884 + 69203296q + \dots$, so that $D(g, s) = 2^{-s}$ tests the m -dependence.

check	form, s	result
Lipschitz, Lemma 5	pointwise, 3 values of s	error $\leq 4 \cdot 10^{-31}$
$L^* = \Lambda(G_k, s)$ on M_k	G_4, G_{12}	error $\leq 10^{-33}$
Theorem 6	$\Delta'; s = 3.3, 5 + 1.5i, 0.37$	error $\leq 8 \cdot 10^{-31}$
Theorem 6	g ; same s	error $\leq 2 \cdot 10^{-26}$
Corollary 7(ii), closed form	$\Delta', g; s = 1, \dots, 11$	exactly 0
Corollary 7(ii), direct	$\Delta', g; s = 3, 6, 9$	$ L_+^* - L_-^* \leq 2 \cdot 10^{-31}$
correction at $s \rightarrow 0$	Δ'	$-2\pi i$ (nonzero)
Prop. 9 (mpmath limits $z \rightarrow -2\pi m \pm i0$)	$\Delta'; s = 3.3, 5 + 1.5i, 0.37$	error $\leq 2 \cdot 10^{-28}$
Prop. 9, wrong side (control)	same	off by 0.012, 0.17, 4.4
Prop. 9	g ; same s	error $\leq 7 \cdot 10^{-26}$
L_+^* vs BFK principal branch	$\Delta'; s = 3.3, 5 + 1.5i$	error $\leq 2 \cdot 10^{-28}$
$L_-^* = \overline{L_+^*}$, real s	$\Delta', g; s = 3.3, 0.37$	error $\leq 10^{-25}$
mean of sides of $\Gamma(0, -2\pi)$ vs $-\text{Ei}(2\pi)$	—	equal to working precision
L_{sym}^* formula vs $\frac{1}{2}(L_+^* + L_-^*)$	$\Delta'; s = 3.3, 5 + 1.5i, -0.7$	error $\leq 2 \cdot 10^{-28}$
functional equation, $K_\pm$	$\Delta', g; s = 3.3, 5 + 1.5i$	error $\leq 10^{-25}$
functional equation, mixed kernel	Δ' / g ; same s	fails by 0.014, 0.38 / 0.001, 0.006
Lemma 3 (ray truncated at 400 periods)	$\Delta', s = -1.5$	agreement $1.7 \cdot 10^{-5}$ (truncation)

Table 1: Sample values: $L_\pm^*(\Delta', 3.3) = -4.269149131848 \dots \pm 0.003080441651 \dots i$ and $L_\pm^*(g, 3.3) = 117227.445841699 \dots \pm 0.000507266006 \dots i$.

10 Status and caveats

Remark 13 (Interior poles). For f with poles in $\mathbb{H}$, Lemma 3 holds for lines avoiding the poles. In the proof of Theorem 6, the circle $|q| = e^{-2\pi y_0}$ now also encloses the images of poles lying *above* height y_0 . The difference then acquires their residues and depends on the height, consistent with the homotopy classes in the draft. This has not been checked numerically.

- *What is classical.* The Lipschitz formula, Hankel's loop, the branch jump of $\Gamma(s, z)$, the conjugacy of lateral values and the principal value, and the natural boundary are all standard. That the negative-index BFK terms need a branch is implicit in [1] and explicit in [2].

- *What is not claimed.* No new term, invariant or L -function. The “principal-part correction” is the summed branch jump (Proposition 9), and its reality-restoring average is the principal value.
- *What the note contributes* (not yet checked exhaustively against the literature). First, an explicit accounting of the choice. Second, the dictionary: side of the cut of $\Gamma(s, \cdot)$ = one-sided inversion of $1 - T$ = direction of the horizontal Mellin ray = choice of coboundary for the T -component of the cocycle. Third, the residue formula at $q = 0$. Fourth, the admissible-direction lemma, which explains why the geometry offers exactly two such choices. Fifth, the fact that the functional equation forces the same side at both cusps. Sixth, the observation that principal-branch values are non-real at real s . The second item is the one with structural content: it connects the analytic continuation over polar modes to the coboundary freedom on the period-polynomial side.
- *Open.* (1) Is there a canonical section, both for the subtraction in the period theory (Section 8) and for the side in $\lambda K_+ + (1 - \lambda)K_-$? Candidates are Hecke (which, by formal bookkeeping, does not distinguish sides), a converse theorem as in [2], the cocycle side, and reality (which selects $\lambda = \frac{1}{2}$). (2) Check Remark 13 and the two remarks of Section 8 numerically on a form with a CM-point pole. (3) Literature: do [1, 2] or later work remark on the non-reality?

References

- [1] K. Bringmann, K.-H. Fricke, Z. Kent, *Special L -values and periods of weakly holomorphic modular forms*, Proc. AMS 142 (2014).
- [2] N. Diamantis, M. Lee, W. Raji, L. Rolin, *L -series of harmonic Maass forms and a summation formula for harmonic lifts*, IMRN 2023(18); arXiv:2107.12366.
- [3] F. Brown, *A class of non-holomorphic modular forms III: real analytic cusp forms for $\mathrm{SL}_2(\mathbb{Z})$* , arXiv:1710.07912.
- [4] Project draft, *Finite-contour cocycles* (Hurwitz T -kernel, Lemma 4.1).